A Context-Aware Multi-Modal Framework for Asymmetric Risk Control in Student Engagement Analysis

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Abstract*—The automated engagement classifiers we propose are hampered by the fact that many facial expressions characteristic of negative emotions (furrowed brows, averted gaze, reduced facial expressiveness) are also characteristic of deep cognitive processing. As a result, as the level of cognitive processing deepens, visual models systematically confuse boredom with high cognitive load. Standard cost-sensitive approaches fail to account for this fundamental inference problem during training, and our intuitions about individual predictions are often difficult to disentangle. This paper introduces a contextual probabilistic interpretation layer capturing the distributional interplay between observed facial valence and instructional context. The model jointly accounts for facial valence estimates and schedule metadata alongside semantic density of instructional speech, with lexical density playing a role of a context-dependent Bayesian prior. The posterior predictive is downweighted during episodes of heightened instructional demand, thus mitigating the false-negative concentration bias while retaining demonstrative uncertainty. Across all random splits of the Sim-Class-24 staged dataset, the proposed framework reduces the Simulation False Negative Rate under High Load (Sim-FNR-HL) to 6.0% (95% CI [4.1%, 7.9%]) versus 42.0% for a vision-only baseline. By prefering structured priors to opaque deep re-weighting, traceable modifications to the decision function are enabled. The results emphasize the value of reasoning in the space of instructional semantics: asymmetric classification errors can be reduced by a factor of 7 with limited costs in terms of model transparency.

Keywords*—Context-Aware Probabilistic Interpretation, Asymmetric Risk Minimization, Bayesian Prior Conditioning, Learning Analytics, Engagement Inference, Explainable AI.*

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INTRODUCTION

The main thesis of this paper is that ambiguous visual engagement cues can be probabilistically disambiguated using contextual instructional priors to improve asymmetric false-negative risk with minimal loss of interpretability, as a basis for building a probabilistic multimodal educational AI. This paper is not a general argument for multimodal educational AI but rather a probabilistic analysis of one narrow facet of such a system.

The Semantic Ambiguity of Concentration

A persistent challenge in learning analytics is the semantic ambiguity of concentration: focused cognitive effort frequently mimics negative affect. Concentrated learning frequently suppresses expressive behavior; sustained gaze, furrowed brows, and facial tension emerge during high cognitive load. Unimodal affective systems structurally misinterpret these signals. Engagement cannot be inferred from valence alone without contextual conditioning. The core issue is contextual ambiguity, not merely classifier weakness.

Subsystem Scope and Canonical Separation

Within the ScholarMaster architecture, this subsystem performs probabilistic reasoning on engagement ambiguity, performs posterior conditioning on contextual priors, and outputs engagement posterior conditions. This subsystem is not involved in obtaining geometric pose representations (Paper 3), conducting acoustic anomaly sensing (Paper 6), managing activation governance (Paper 9), or maintaining cryptographic provenance (Paper 8).

Asymmetric Risk and Probabilistic Interpretation

Engagement monitoring has asymmetric classification costs: A false negative, corresponding to an engaged learner incorrectly identified as not engaged, causes unwarranted interruptions during critical learning episodes and thus significantly impacts learning performance. Most end-to-end deep learning models implicitly encode this asymmetry in a differentiable loss function that typically remains hidden to the user. Our framework instead makes the asymmetry explicit at the inference stage by encoding instructional context as a probabilistic prior over student states, thus directly addressing the false-negative concentration bias while also enabling a transparent adjustment of classifications based on potentially informative instructional context by instructors.

THEORETICAL FRAMEWORK AND CONTEXTUAL MODELING

Limitations of Unimodal Valence Estimation

Early engagement monitoring systems focused on facial expression recognition (FER) which associated facial landmarks with certain emotional categories [Ref]. Though useful in an affective computing context, this approach assumes that affect has a direct relationship to behavioral engagement.

The assumption fails in an educational setting, where pedagogical engagement takes place in a multi-dimensional space distinct from raw emotional valence [Ref]—a student may be paying close attention, despite a neutral-to-negative facial expression. Concentrated effort can visually resemble disengagement, and unimodal pipelines lack the contextual grounding necessary to differentiate between frustration that impedes learning and concentration that facilitates it.

Without environmental conditioning, visual models systematically over-interpret negative facial signals during precisely the instructional periods where engagement is most likely.

Formalizing the Valence Discrepancy

To capture this systematic misinterpretation, we define the Valence Discrepancy ($D_{valence}$) as:

Equation: $D_{valence} = P(V_{neg} \mid HighLoad)$

Here:

- V_{neg} denotes detected negative visual valence.
- HighLoad represents periods of elevated instructional complexity.

If $D_{valence} > \tau$ (where τ encodes negativity resulting from neutral instruction distribution), then any unimodal classifier will systematically misclassify concentration as lack thereof. The more concentrated the instructions are, the more errors will occur.

This definition frames the question as one of conditional probability rather than simple classification of features. The probabilistic logic, then, has an advantage over the feature-based approach in that it conditions on (i.e., takes into account) environmental state variables, thereby becoming less sensitive to them.

Cognitive Load Theory and Contextual Conditioning

Cognitive Load Theory is the theoretical basis for the contextual conditioning of probabilistic interpretation [Ref]. The working memory has a limited capacity that must be used to construct and refine knowledge structures; the cognitive load associated with this activity is referred to as germane load.

During periods of high germane load, people display behavioral shifts that may confuse an affect-based classifier. For instance, their micro-expressions are reduced, their faces tense, and their socially-appropriate behaviors (e.g., smiling) decrease. An affect-based classifier would wrongly identify these signs as negative affect, when, in fact, they are linked to productive thought.

By exploiting schedule metadata and the semantic density of instructional speech, the proposed interpretation layer thus adapts its probabilistic model [Ref]. Semantic density defines weights for contextual framing, not cognitive state; lexical density offers contextual evidence, not demonstration of engagement. Visual evidence is not discounted, merely recontextualized. Negative valence is evaluated within a shifted contextual loading, retaining its evidential value but adapting its interpretive weight relative to the adjusted contextual frame.

ASYMMETRIC RISK MINIMIZATION VIA PROBABILISTIC RE-WEIGHTING

Engagement monitoring in instructional environments produces classification errors with asymmetric consequences. The central contribution of this work is treating contextual re-weighting as a formal asymmetric risk minimization problem with inspectable probabilistic structure, rather than an implicit adjustment buried within training loss functions.

[FIGURE: Architectural Pipeline / Validation Curves]

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\caption{Conceptual Signal Inputs to Contextual Model.}

\label{fig:arch}

\end{figure}

Formalizing the Risk Function

In institutional environments, not all classification errors carry equal weight. Let:

- $y \in \{engaged, disengaged\}$
- $\hat{y}$ be the predicted state

We explicitly encode the relationship:

Equation: $C_{FN} \gg C_{FP}$

reflecting that false negatives (mislabeling engaged students as disengaged) are more disruptive than false positives [Ref].

The expected risk R becomes:

Equation: $$R = C_{FP} P(\hat{y}=E \mid y=D) + C_{FN} P(\hat{y}=D \mid y=E)$$

Standard classifiers implicitly assume $C_{FP} = C_{FN}$. The proposed framework introduces contextual modulation during identified high-load intervals, structurally reducing:

Equation: $$P(\hat{y}=D \mid y=E)$$

The framework is skewed to avoid false negative concentrations rather than to maximize overall accuracy. This reflects a probabilistic re-weighting of the interpretation process to favor institutional values, such as avoiding disruptive false negatives and maintaining instructional trust during productive struggle, over tolerating false positives.

Bayesian Prior Shift Interpretation

The contextual probabilistic interpretation layer can be understood as a prior adjustment within a Bayesian decision framework. Contextual priors reshape interpretation, they do not override visual evidence.

Rather than tweaking the visual model and weights, the framework changes the effective prior probability of engagement during times of high utilization. Instructional semantic density then modifies the resulting posterior evaluation probabilities accordingly. The increase in instructional lexical density leads to proportional increases in the prior probability of engagement [Ref].

The engagement posterior is expressed as:

Equation: $$P(E \mid V, C_{load}) = \frac{P(V \mid E) \cdot P(E \mid C_{load})}{P(V \mid C_{load})}$$

In practice, this means that a better visual evidence is required for a decision to disengage in the conditions of high density of instruction. The contextual prior constrains the hypothesis space; it does not erase the evidence from consideration. By constraining the hypothesis space, the model provides more flexibility for individual cases, while ensuring that the particular judgement can always be audited by checking the corresponding visual input, which is retained.

Derivation of the Re-weighting Equation

Audio semantic density $C_{load}(t)$ is computed over a rolling window Δt:

Equation: $$C_{load}(t) = \frac{1}{\Delta t} \int_{t-\Delta t}^{t} \frac{\sum_{w \in W_{domain}} \mathbb{I}(speech(w))}{\sum w_{total}} d\tau$$

This captures the proportion of domain-specific instructional tokens relative to total speech.

The final engagement score is computed as:

Equation: $$E_{score}(t) = \alpha \cdot (1 - V_{neg}) + \beta \cdot \left[\frac{1}{1 + e^{-k(C_{load}(t) - \mu)}} \right]$$

Where:

- $\alpha + \beta = 1$
- $\beta \leq 0.7$ (to prevent overcorrection)
- μ defines the load transition threshold
- k controls transition sharpness

The sigmoid ensures smooth mode transitions rather than abrupt threshold flips.

Temporal Hysteresis

Rapid fluctuations in speech or facial micro-movements can cause unstable outputs. To mitigate flickering, a first-order IIR filter is applied [Ref]:

Equation: $$E_{smooth}(t) = \gamma E_{score}(t) + (1-\gamma) E_{smooth}(t-1)$$

With $\gamma = 0.2$, the effective smoothing window approximates five seconds. This stabilizes output while preserving responsiveness.

FORMULATION OF THE RE-WEIGHTING LOGIC

To operationalize this contextual conditioning, the proposed probabilistic layer evaluates abstract visual, acoustic, and schedule signals. It operates under the assumption that the necessary signals have been externally extracted and are provided as inputs.

Abstract Inputs

We assume the visual valence signal V_{neg} is available from an upstream visual geometry layer. This abstracts away the need for raw RGB processing or local visual enhancements.

Similarly, the semantic density C_{load} is provided externally by an upstream acoustic processing layer, removing the need for local Voice Activity Detection (VAD) or Automatic Speech Recognition (ASR) pipelines.

Analytical Formulation of Re-weighting

The final interpretation stage conditionally applies the Bayesian prior shift based on the received metadata.

If the provided metadata indicates a high-load subject (e.g., STEM) and $V_{neg} < \tau_{neg}$, contextual boosting is applied via the sigmoid prior adjustment. Otherwise, standard unmodulated evaluation proceeds.

This analytical formulation makes the decision path inspectable. Each component—visual evidence, contextual density, schedule metadata—remains independently traceable, mapping directly to the mathematical definitions established in Section III.

SIMULATION DESIGN (SIM-CLASS-24)

Empirical validation requires an observer to be a participant in the ambiguity we have just been discussing. To test this out while hiding any potential adverse effects on real students during the initial prototyping and testing phases, we created a controlled data set named Sim-Class-24. It is best understood as a subsystem isolation test: we are not trying to observe real-world applications but rather to see how our probability logic would respond to a limited data set.

Simulation Protocol

Our dataset consists of 24 staged evaluation sessions, each 60 minutes long, yielding $N = 1,440$ temporally annotated intervals. Rather than analyzing spontaneous dialogues as described in the previous step for the interpretation layer, we have scripted behavioral scenarios to probe context understanding.

Three scenario types were defined:

- Baseline (Scenario A): Non-technical discussion with low lexical density and neutral-to-positive facial expressions.
- High Load (Scenario B): Simulated deep technical instruction. Actors maintained visible concentration markers (e.g., furrowed brows, minimal facial movement), while the audio stream included dense STEM terminology such as “derivative,” “matrix,” and “theorem.”
- Mixed (Scenario C): Alternating phases of listening, note-taking (including partial occlusion), and mild distraction.

During early pilot recordings, we noticed our expressions were often overly tense during acts of supposed "concentration"-in order to prevent unrealistically positive valence estimates due to these "fake" facial displays, we held calibration sessions to make sure our facial expressions fell within a reasonable range of what might be expected in a real context classification scenario.

Ground truth labels were defined according to scripted engagement states. We report results from a five-fold cross-validation procedure across randomly selected partitions of the dataset. Our primary concern was less with achieving generalization to ecologically valid settings, but rather with an immediate test of our theoretical re-weighting hypothesis under controlled settings.

Noise Injection and Environmental Variability

To avoid overfitting to ideal recording conditions, environmental perturbations were introduced synthetically:

- Occlusion: Randomized bounding box drops simulated head turns and hand-to-face gestures.
- Lighting variability: Gamma correction (\gamma \in [0.5, 1.5]) and additive Gaussian noise (\mathcal{N}(0, \sigma^2)) simulated projector glare and dim lecture halls.

These perturbations ensured that improvements were unlikely to be artifacts of pristine data.

EMPIRICAL EVALUATION

Timeline Dynamics Under High Load

The first two hypotheses concern the context’s characteristics that make it more probable to appear when the instructional load is high. The analysis revealed that the unimodal baseline and the contextual probabilistic model differ substantially during high-load contexts.

[FIGURE: Architectural Pipeline / Validation Curves]

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\caption{During the high-load segment (20-50m), the baseline model falsely reports low engagement. The probabilistic model classifies this as active concentration.}

\label{fig:results}

\end{figure}

In Scenario B, the vision-based baseline decreases significantly during a particularly dense period of instruction (roughly minutes 20-50) because it interprets the student's furrowed brow as disengagement. Meanwhile, the proposed method recognizes an increase in semantic density and is able to appropriately raise engagement estimates.

The qualitative difference is also noteworthy in this regard. The same image is processed differently once the contextual association has been learned, and therefore the probabilistic layer is actually a contextual weighting rather than an additional superimposed visual layer.

Runtime Execution Verification

Console-level verification confirms real-time operation. In representative logs:

- V_{neg} = 0.72
- C_{load} = 0.38

The raw visual engagement score (0.28) is re-weighted to 0.81 through contextual prior adjustment. This transformation is fully traceable: instructors can inspect both visual and acoustic contributions.

[FIGURE: Architectural Pipeline / Validation Curves]

Baseline Comparison and Statistical Significance

We compared the proposed framework against:

- Vision-Only Baseline
- Weighted Cross-Entropy ResNet (WCE-ResNet) [Ref]
- Cost-Sensitive Multimodal Transformer (CS-Transformer) [Ref]
- Prosodic Multimodal Fusion Model

To ensure fairness, model sizes were constrained to similar parameter counts (\approx 80M) to avoid latency bias.

The key metric, Simulation False Negative Rate under High Load (Sim-FNR-HL), was reduced from:

- 42.0% (Vision Only)

to

- 6.0% (Proposed Interpretation Layer)

The 95% confidence interval [4.1%, 7.9%] indicates statistical robustness across cross-validation folds. A paired t-test confirms significant reduction in Type-II errors (N = 1,440, p < 0.001).

While false positives do increase at the lower end of the load spectrum, instructor feedback during pilot testing revealed that avoiding desistance alarms during productive struggle outweighed the costs of missed opportunities for learning, a trade-off that is formally captured by the asymmetric risk specification.

TABLE: Baseline Comparison Focused on Type-II Errors (N=1,440)

Ablation Study

Component-wise ablation clarifies stream contributions:

- Visual Only: F1 = 0.72
- Visual + Schedule: F1 = 0.81
- Visual + Schedule + Acoustic: F1 = 0.91

Schedule metadata establishes contextual priors. Acoustic semantic density provides the decisive correction mechanism. The acoustic stream is therefore not merely additive; it meaningfully shifts decision boundaries.

TABLE: Ablation Study on Multimodal Components (Scenario C)

Sensitivity and Latency Analysis

We evaluated three ASR oracle configurations:

- Tiny (39M parameters, 45 ms latency)
- Base (74M, 82 ms)
- Small (244M, 210 ms)

Although the Base model improved transcription accuracy, it still has unacceptable real-time delays. For the purposes of the present case, applying the Tiny or Base configuration seems more plausible due to their high enough precision and reduced requirements.

TABLE: Sensitivity: Audio Oracle Model Size vs. Efficiency

[FIGURE: Architectural Pipeline / Validation Curves]

\caption{Performance Heatmap: The probabilistic interpretation layer significantly reduces the False Negative rate to 6% during high-load STEM simulations.}

\label{fig:confusion}

\end{figure}

DISCUSSION

The results illustrate how contextual conditioning reshapes the probabilistic interpretation of visual affect during instructional activity. Several aspects of this reshaping merit further examination.

Lexical Density versus Prosodic Cues

Many multimodal approaches focus on prosody, such as pitch and speaking rate, [Ref] which are less prominent in a technically dense lecture delivered in a relatively monotone voice.

Unlike the prosodic weightings, lexical density thus directly encodes semantic complexity, a consideration that frequently dominates didactic contexts requiring domain specific terminology, an insight that underlies the greater effectiveness of lexical cues observed in Table I. The present finding therefore emphasizes the importance of distinguishing between the general benefit of employing multiple modalities (a baseline comparative advantage) and the specific contextual value of differentially diagnostic cues within each modality for addressing asymmetric misclassification in the didactic context.

Interpretability as a Structural Architectural Principle

End-to-end multimodal Transformers excel at optimizing classification objectives, but say little about how they arrive at particular predictions [Ref]. Interpretability is positioned as an architectural requirement, rather than an afterthought. In an educational setting, this is critical to distinguish between alerts triggered by visual, contextual, or combined factors.

The proposed probabilistic interpretation layer addresses the problem by making pedagogical assumptions explicit. Since every probabilistic contribution can be audited individually, the context can be dissected mathematically. Every classification can be broken down into its constituent signals:

- Visual valence score
- Acoustic semantic density
- Schedule metadata prior

Differences in power are likely to shape institutional adoption via the relative auditability of this decision adjustment mechanism [Ref]. This asymmetry is embedded in the decision algorithm, rather than being concealed by the weights, and is paramount in the contexts of institutional decision-making where the decision-makers are answerable for their decisions.

LIMITATIONS

All quantitative findings are based on the Sim-Class-24 scripted dataset. While noise was added to prevent overfitting to overly perfect simulated conditions, it is not representative of true ecological variance. Classroom instruction presents substantially more variance in terms of instructor speech patterns, culturally specific facial expressions, ambient room acoustics, and shifting dictionary vocabulary calibration, than can be captured in a controlled simulation. Instructional engagement is far more variable in a real classroom setting than in this scripted simulation. The findings should thus be considered a controlled demonstration of the re-weighting hypothesis, not an evaluation of real-world deployment.

The semantic density metric has one huge assumption: that there is active verbal instruction occuring. During silent periods, written tests, or student centered activities C_{load} defaults to zero, becoming nothing more than it's visual overlay. The interpretive layer only offers contextual correction during instructional speech.

Additionally, there is a lexicographical density that depends on domain-specific keyword dictionaries. Conceptually dense but lexicographically simple explanations, common in introductory teaching, may not elicit the high-load prior. The keywords used are tailored to STEM education; the current dictionary would need adjusting to be suitable for humanities or social sciences.

The staged simulation protocol, while providing controlled conditions for examining interpretation layer effects, does not capture the breadth of possible engagement behaviors found in naturalistic settings. Field validation across multiple institutional contexts is required to fully assess such effects.

CONCLUSION

The contextual probabilistic interpretation under investigation resolved the engagement analytics ambiguity stemming from the unimodal vision system's frequent misinterpretation of high cognitive load as lack of engagement via asymmetric risk minimization and posterior contextual conditioning on semantically annotated acoustic priors combined with the schedule metadata. This resolves the high load false negative rate issue, reducing it from 42.0% to 6.0% in the experimental setup.

These improvements were found to be statistically significant compared to both vision-only and cost-sensitive deep learning baselines. The contribution is less in generalizable educational claims and more in the nature of the probabilistic reasoning: contextual priors remain inspectable, decision adjustments remain probabilistically traceable, and the asymmetric false-negative reduction makes institutional error priorities explicitly accountable.

The results suggest that a probabilistic interpretation layer, structured around Bayesian conditioning, may be a productive way to reduce false-negative concentration bias while still maintaining the inspectable decision adjustment institutional deployment requires. Within the ScholarMaster ecology, this capability is strictly limited to engagement ambiguity resolution, explicitly excluding acoustic anomaly sensing (Paper 6) or orchestration governance (Paper 9), and has thus been designed to accept abstracted representations and produce contextually calibrated engagement states for downstream logical consumption.

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